LCIM — Logical Consistency Integrity Module

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Independent Methodological Authority

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Governed Space: Logical Consistency Integrity
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Cognitive Reasoning Integrity Standard (CRIS) ·
Inference Integrity Module (IIM) ·
Assumption Integrity Module (AIM) ·
Evidence Evaluation Integrity Module (EEIM) ·
Multi-Layer Truth Validation Framework (MTVF) ·
INTEGROS® — Integrity Standard
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Canonical Definition

Logical Consistency Integrity Module (LCIM) defines the structural conditions under which reasoning processes, logical relationships, conclusions, assumptions, evidence dependencies, and cognition outputs remain internally coherent, contradiction-aware, structurally compatible, and governance-valid throughout reasoning activities.

LCIM governs logical consistency.

The module ensures that cognition remains capable of maintaining coherent reasoning structures without generating unresolved contradictions, incompatible conclusions, self-invalidating logic, or governance-invalid reasoning states.

A system satisfies LCIM only if:

- logical consistency remains assessable
- contradictions remain detectable
- incompatible conclusions remain identifiable

- reasoning coherence remains preservable
- logic conflicts remain governable
- cognition outputs remain structurally compatible

A cognition environment that cannot identify materially significant contradictions does not satisfy LCIM.

Module Operational Role

LCIM defines the logical-consistency governance layer of CRIS.

Its role is to preserve coherence across reasoning structures.

Module Operational Space

LCIM governs:

- logical consistency
- contradiction detection
- reasoning coherence
- conclusion compatibility
- logic integrity
- structural consistency
- cognition coherence
- contradiction governance

The module applies wherever reasoning generates multiple conclusions, relationships, assumptions, or evidence structures.

Module Function

The module applies wherever systems must preserve:

- coherent reasoning
- contradiction awareness
- governance-valid logic
- structurally compatible conclusions
- reasoning stability
- operationally reliable cognition

Its function is to ensure that cognition remains logically coherent throughout reasoning processes.

Minimum Implementation Framework

1. Define the Logical Consistency Object

The organization must define which reasoning structures require governance.

This may include:

- reasoning chains
- analytical conclusions
- strategic assessments
- operational recommendations
- planning outputs
- evidence relationships

- assumption structures
- autonomous reasoning outputs

2. Define Logical Consistency Conditions

The system must define the conditions under which logical consistency remains valid.

This includes:

- contradiction-management requirements
- coherence requirements
- compatibility requirements
- structural-integrity requirements
- reasoning-validity requirements
- governance-valid logic conditions

3. Define Logical Consistency Degradation Detection Logic

The system must define how logical-consistency degradation is identified.

This may include:

- contradictory conclusions
- incompatible assumptions
- logic conflicts
- coherence failures
- reasoning inconsistency
- structural incompatibility

4. Define Operational Response or Governance Logic

The system must define governance logic for logical-consistency failures.

Governance response may include:

- contradiction review
- coherence validation
- logic reconstruction
- governance intervention
- escalation
- conclusion restriction
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- contradiction events
- logic reviews
- consistency validations
- governance actions
- reasoning structures
- resulting cognition outputs

A cognition environment must not remain logic-valid if materially significant contradictions cannot be identified, reviewed, or governed.

Use Case 1 — Autonomous Policy Analysis Agent

Scenario

An autonomous policy-analysis system evaluates large volumes of regulations, requirements, and organizational policies.

Application

LCIM governs contradiction detection, conclusion compatibility,
and logical coherence throughout policy reasoning.

Result

The organization gains stronger consistency assurance, reduced reasoning conflicts,
and improved governance reliability.

Use Case 2 — Strategic Planning Environment

Scenario

A strategic planning system continuously generates recommendations
regarding future organizational actions and priorities.

Application

LCIM governs logical consistency between assumptions, evidence, objectives,
and resulting conclusions.

Result

The environment gains stronger planning coherence, improved reasoning stability,
and reduced exposure to contradictory outcomes.

Canonical Closing Statement

Logical Consistency Integrity Module (LCIM) defines the structural conditions
under which reasoning processes, logical relationships, conclusions, assumptions,
evidence dependencies, and cognition outputs remain internally coherent,
contradiction-aware, structurally compatible, and governance-valid
throughout reasoning activities.

Reasoning cannot remain valid when contradictions become invisible.
Logical consistency therefore becomes a foundational integrity condition
of governable reasoning.

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Canonical Source

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